

Pushing the Selection–Plan 3 Hybrid

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1 Purpose

This note tries to prove the hybrid replacement for Plan 1 graph entry: use selection to get a controlled class, go to a Plan 3 good level, upgrade D_H to L^2 -Serrin and then to L^∞ -Serrin, apply a Brandolini-type annulus theorem, and conclude graph entry.

The positive result is a rigorous quantitative interpolation lemma in dimension two. The scale tradeoff is: the layer depth needed to get L^∞ -Serrin smallness is generally too large to preserve the selected asymmetry unless one knows the stronger relation

$$\delta_T(U) \ll \alpha(U)^2.$$

This first looked like an obstruction. But in a contradiction proof of $\alpha(U) \lesssim \delta_T(U)^{1/2}$, the bad alternative is exactly $\alpha(U) \gg \delta_T(U)^{1/2}$, hence exactly $\delta_T(U) \ll \alpha(U)^2$. Thus the scale tradeoff is not the remaining obstruction. The remaining obstruction is the geometric entry theorem: L^∞ -Serrin smallness on the chosen level must put that level into the small radial-graph class, without using BDV as a black box.

2 Standing Optimistic Hypotheses

Let $U \subset B_R \subset \mathbb{R}^2$ be a selected minimizer, normalized near the unit disk, and let u be its torsion function. Write

$$\delta := \delta_T(U), \quad a := \alpha(U), \quad q := \delta/a.$$

To give the hybrid the best possible chance, assume the following uniform geometric estimates, all with constants depending only on the selected class:

(H1) **Lipschitz potential and linear boundary growth.** There are $0 < c_0 < L_0 < \infty$ such that near ∂U ,

$$c_0 \operatorname{dist}(x, \partial U) \leq u(x) \leq L_0 \operatorname{dist}(x, \partial U), \quad |\nabla u(x)| \leq L_0. \quad (2.1)$$

(H2) **Tubular layer bound.**

$$|\{x \in U : \operatorname{dist}(x, \partial U) \leq r\}| \leq P_0 r \quad \text{for } 0 < r < r_0. \quad (2.2)$$

(H3) **Level-set lower density.** For every regular level $\Sigma_t = \{u = t\}$ considered below, every $x \in \Sigma_t$, and every $0 < r < r_0$,

$$\mathcal{H}^1(\Sigma_t \cap B_r(x)) \geq c_1 r. \quad (2.3)$$

Remark 2.1. *These hypotheses are stronger than what the selection step alone currently proves before graph entry. They are included to isolate the numerical obstruction even under favorable assumptions.*

Theorem 2.2 (Conditional hybrid estimate). *Assume (H1)–(H3) and the Plan 3 volume-variable identity*

$$\int_0^{|U|} D_H(v(s)) ds \leq C\delta.$$

Then, for every $0 < \lambda < \lambda_0$, there is a regular level $\Sigma_{\hat{t}} = \partial E_{\hat{t}}$ such that

$$\lambda/2 \leq |U \setminus E_{\hat{t}}| \leq \lambda, \quad D_H(\hat{t}) \leq C\delta/\lambda,$$

and

$$\text{dist}(\Sigma_{\hat{t}}, \partial U) \geq c\lambda.$$

Writing $f = |\nabla u|$ on $\Sigma_{\hat{t}}$ and $\bar{f} = |E_{\hat{t}}|/P(E_{\hat{t}})$, one has

$$\int_{\Sigma_{\hat{t}}} (f - \bar{f})^2 d\mathcal{H}^1 \leq C \frac{\delta}{\lambda}$$

and

$$\|f - \bar{f}\|_{L^\infty(\Sigma_{\hat{t}})} \leq C \left[\left(\frac{\delta}{\lambda^2} \right)^{1/3} + \left(\frac{\delta}{\lambda} \right)^{1/2} \right].$$

In particular, for $\delta, \lambda \leq 1$,

$$\|f - \bar{f}\|_{L^\infty(\Sigma_{\hat{t}})} \leq C \left(\frac{\delta}{\lambda^2} \right)^{1/3}.$$

Consequently, this mechanism can simultaneously give fixed small L^∞ -Serrin oscillation and preserve a fixed fraction of the selected asymmetry only in the stronger regime

$$\delta \leq C\alpha(U)^2.$$

This stronger regime is automatically available in the bad alternative of the desired estimate $\alpha(U) \leq C\sqrt{\delta}$.

The proof is the chain of estimates in Sections 3–7.

3 A Good Level in a Prescribed Layer

Let

$$\mu(t) := |U \setminus E_t|, \quad E_t := \{u > t\}.$$

The Plan 3 volume-variable identity gives

$$\int_0^{|U|} D_H(v(s)) ds \leq C\delta$$

in the normalized two-dimensional regime, up to harmless constants. Hence, for any layer depth $0 < \lambda < \lambda_0$, there is a regular level $\hat{t}$ with

$$\lambda/2 \leq \mu(\hat{t}) \leq \lambda, \quad D_H(\hat{t}) \leq C \frac{\delta}{\lambda}. \tag{3.1}$$

Indeed, average over the volume slab $\{s : |U| - \lambda \leq s \leq |U| - \lambda/2\}$, whose length is $\lambda/2$, and intersect with regular values.

The layer bound (2.2) and the lower growth in (2.1) give

$$\mu(t) = |\{0 < u \leq t\}| \leq |\{\text{dist}(x, \partial U) \leq t/c_0\}| \leq Ct.$$

Thus $\mu(\hat{t}) \geq \lambda/2$ implies

$$\hat{t} \geq c\lambda. \quad (3.2)$$

The upper Lipschitz bound in (2.1) then implies every point of $\Sigma_{\hat{t}}$ lies at distance at least

$$d_\lambda := c\lambda \quad (3.3)$$

from ∂U .

4 D_H Gives L^2 -Serrin

On $\Sigma_{\hat{t}}$, set $f = |\nabla u|$ and $\bar{f} = |E_{\hat{t}}|/P(E_{\hat{t}})$. The exact identity gives

$$\int_{\Sigma_{\hat{t}}} \frac{(f - \bar{f})^2}{f} d\mathcal{H}^1 = \frac{|E_{\hat{t}}|}{P(E_{\hat{t}})^2} D_H(\hat{t}).$$

Using $|\nabla u| \leq L_0$ and the uniform perimeter/area bounds in the selected class,

$$\int_{\Sigma_{\hat{t}}} (f - \bar{f})^2 d\mathcal{H}^1 \leq CD_H(\hat{t}) \leq C \frac{\delta}{\lambda}. \quad (4.1)$$

5 Interior Regularity Gives a Lipschitz Modulus on the Level

Since $\Sigma_{\hat{t}}$ is at distance $d_\lambda \geq c\lambda$ from ∂U , interior estimates apply. Moreover, by the upper growth in (2.1), in balls of radius comparable to d_λ around $\Sigma_{\hat{t}}$ one has $\sup u \leq Cd_\lambda$. Standard interior estimates for $-\Delta u = 1$ therefore give

$$\|D^2 u\|_{L^\infty(\text{a } c\lambda\text{-collar of } \Sigma_{\hat{t}})} \leq \frac{C}{\lambda}. \quad (5.1)$$

Consequently

$$\text{Lip}_{\Sigma_{\hat{t}}}(|\nabla u|) \leq \frac{C}{\lambda}. \quad (5.2)$$

6 The $L^2 \rightarrow L^\infty$ Upgrade in Dimension Two

Lemma 6.1 (Level interpolation). *Let Σ be a regular level satisfying the lower density (2.3). If g is Lipschitz on Σ , with*

$$\int_{\Sigma} g^2 d\mathcal{H}^1 \leq B, \quad \text{Lip}_{\Sigma}(g) \leq M,$$

then

$$\|g\|_{L^\infty(\Sigma)} \leq C(c_1, r_0) \left((BM)^{1/3} + B^{1/2} \right).$$

In particular, if $0 < B \leq B_0$, $0 < M$, and $BM \geq B^{3/2}$, then

$$\|g\|_{L^\infty(\Sigma)} \leq C(c_1, r_0, B_0) (BM)^{1/3}.$$

Proof. Let $A = \|g\|_{L^\infty(\Sigma)}$, attained up to approximation at a point $x \in \Sigma$. If $M = 0$, then g is constant on each connected component of Σ , and the lower density at one point gives $B \geq c_1 r_0 A^2$ after replacing r_0 by a smaller dimensional radius if necessary. Hence $A \leq CB^{1/2}$.

Assume now $M > 0$, and set $r = A/(2M)$. If $r \leq r_0$, then $|g| \geq A/2$ on $\Sigma \cap B_r(x)$, and (2.3) gives

$$B \geq \int_{\Sigma \cap B_r(x)} g^2 d\mathcal{H}^1 \geq \frac{A^2}{4} c_1 r = c \frac{A^3}{M}.$$

Hence $A \leq C(BM)^{1/3}$. If $r > r_0$, then the same argument with r_0 gives $B \geq cA^2 r_0$, hence $A \leq CB^{1/2}$. $\square$

Apply Lemma 6.1 to

$$g = f - \bar{f}.$$

By (4.1) and (5.2),

$$B \leq C \frac{\delta}{\lambda}, \quad M \leq C \frac{1}{\lambda}.$$

Therefore

$$\| |\nabla u| - \bar{f} \|_{L^\infty(\Sigma_i)} \leq C \left[\left(\frac{\delta}{\lambda^2} \right)^{1/3} + \left(\frac{\delta}{\lambda} \right)^{1/2} \right]. \quad (6.1a)$$

In the regime relevant to near-balls, $\delta, \lambda \leq 1$, the second term is dominated by the first:

$$\left(\frac{\delta}{\lambda} \right)^{1/2} \leq \left(\frac{\delta}{\lambda^2} \right)^{1/3}.$$

Thus

$$\| |\nabla u| - \bar{f} \|_{L^\infty(\Sigma_i)} \leq C \left(\frac{\delta}{\lambda^2} \right)^{1/3}. \quad (6.1b)$$

This is the strongest rigorous estimate obtainable from the proposed mechanism, even under the optimistic hypotheses above.

7 The Layer-Depth Tradeoff

To make the L^∞ -Serrin oscillation smaller than a prescribed ε_B , (6.1b) requires

$$\lambda \geq C \varepsilon_B^{-3/2} \sqrt{\delta}. \quad (7.1)$$

This is exactly the $\sqrt{\delta}$ -scale found in the Plan 3 push.

But preserving the selected asymmetry requires a much smaller layer. Since α is L^1 -Lipschitz on bounded sets,

$$\alpha(E_i^\sharp) \geq \alpha(U) - C|U \setminus E_i| \geq a - C\lambda. \quad (7.2)$$

Thus to keep, say,

$$\alpha(E_i^\sharp) \geq \frac{1}{2} \alpha(U),$$

one needs

$$\lambda \leq ca. \quad (7.3)$$

Combining (7.1) and (7.3), the route can simultaneously produce L^∞ -Serrin smallness and preserve asymmetry only if

$$\sqrt{\delta} \leq Ca, \quad \text{equivalently} \quad \delta \leq Ca^2. \quad (7.4)$$

In terms of $q = \delta/a$, this is

$$q \leq Ca. \quad (7.5)$$

8 The Correct Contradiction Split

The desired conclusion is

$$a = \alpha(U) \leq C_* \sqrt{\delta}.$$

Fix a large constant M , to be chosen from the constants below. If

$$a \leq M\sqrt{\delta},$$

there is nothing to prove. Otherwise assume

$$a > M\sqrt{\delta}. \quad (8.1)$$

Choose

$$\lambda = K\sqrt{\delta}, \quad (8.2)$$

with K large enough that (6.1b) gives

$$\| |\nabla u| - \bar{f} \|_{L^\infty(\partial E_t)} \leq \varepsilon_B. \quad (8.3)$$

Indeed

$$\left(\frac{\delta}{\lambda^2} \right)^{1/3} = K^{-2/3}.$$

Thus $K \geq C\varepsilon_B^{-3/2}$ suffices.

The same choice is still small relative to the selected asymmetry in the bad alternative: if $M \geq 4CK$, then

$$\lambda = K\sqrt{\delta} \leq \frac{K}{M}a \leq \frac{a}{4C},$$

and therefore

$$\alpha(E_t^\sharp) \geq a - C\lambda \geq \frac{3}{4}a. \quad (8.4)$$

This lower bound is not actually needed for the final transfer, but it shows that the selected badness has not been destroyed by the inset.

The Plan 3 profile transfer for superlevel sets gives, for every small λ and not only for $\lambda = O(\delta)$,

$$\delta_T(E_t) \leq \frac{1}{1 - \lambda/|U|} \delta_T(U) \leq 2\delta \quad (8.5)$$

after taking δ small enough that $\lambda \leq |U|/2$. This is the exact profile computation using $u - \hat{t} = u_{E_t}$; it is not a monotonicity assertion.

Now assume the remaining geometric entry statement:

If a regular superlevel set $E_{\hat{t}}$ in the selected planar class satisfies (8.3), then after translation and volume normalization it belongs to the small-amplitude radial graph class to which the direct graph theorem applies.

Then the direct radial-graph theorem gives

$$\alpha(E_{\hat{t}})^2 \leq C_G \delta_T(E_{\hat{t}}) \leq C'_G \delta. \quad (8.6)$$

Finally the elementary superlevel transfer gives

$$\alpha(U) \leq \alpha(E_{\hat{t}}) + C|U \setminus E_{\hat{t}}| \leq C\sqrt{\delta} + CK\sqrt{\delta} \leq C_*\sqrt{\delta}. \quad (8.7)$$

Choosing $M > C_*$ contradicts the bad alternative (8.1). Therefore $a \leq M\sqrt{\delta}$.

9 Checking the Selected-Regularity Route to Brandolini

The geometric entry statement in the previous section is supplied if the selection step is used with its full regularity output, not merely with the coarse Lipschitz/density bounds (H1)–(H3). The needed input is:

(SR) The selected planar minimizer U has a uniformly controlled $C^{2,\alpha}$ boundary in a fixed atlas, the torsion function has a uniform $C^{2,\alpha}$ norm up to the boundary, and

$$0 < q_- \leq |\nabla u| \leq q_+ \quad \text{on } \partial U, \quad (9.1)$$

with all constants depending only on the selected class.

This is exactly the kind of output provided by the Plan 1 graph-entry plus Schauder bootstrap package: after the selected set is known to be a $C^{1,\gamma}$ graph, the selected Bernoulli law and the uniform Schauder bootstrap give $C^{m,\gamma}$ bounds, in particular $C^{2,\alpha}$, with constants depending only on the selected class.

Lemma 9.1 (Inset regularity inherited from the selected boundary). *Assume (SR). Then there are $t_0 > 0$ and constants ρ_0, K_0, d_0 , depending only on (SR), such that every regular level $E_t = \{u > t\}$, $0 < t < t_0$, is connected and has $C^{2,\alpha}$ boundary in the sense of Brandolini's Definition 6, with atlas constants (ρ_0, K_0) and diameter at most d_0 .*

Proof. By (9.1) and the C^1 -continuity of ∇u up to ∂U , after shrinking a fixed tubular collar one has

$$|\nabla u| \geq q_-/2. \quad (9.2)$$

In normal coordinates $x = y - s\nu_U(y)$, $y \in \partial U$, the map $(y, s) \mapsto u(y - s\nu_U(y))$ is $C^{2,\alpha}$, has $\partial_s u \leq -q_-/2$, and has uniformly bounded $C^{2,\alpha}$ norm. The quantitative implicit function theorem therefore gives, for $0 < t < t_0$, a $C^{2,\alpha}$ function $s_t(y)$ with uniformly bounded $C^{2,\alpha}$ norm such that

$$\partial E_t = \{y - s_t(y)\nu_U(y) : y \in \partial U\}. \quad (9.3)$$

The uniform atlas constants follow from the corresponding constants for ∂U , the tubular radius, and the implicit-function constants. The small selected graph class contains a fixed inner ball, and the linear boundary growth gives E_t a slightly smaller fixed inner ball for $t < t_0$; hence these near-boundary levels are connected. The diameter bound is inherited from $U \subset B_R$. $\square$

Lemma 9.2 (Brandolini annulus for the stopped level). *Assume (SR) and choose the stopped level in the bad alternative with $\lambda = K\sqrt{\delta}$. After volume normalization of $E_{\hat{t}}$, there are a center z and a modulus $\eta(K, \delta)$ such that*

$$B_{1-\eta(K, \delta)}(z) \subset E_{\hat{t}}^{\sharp} \subset B_{1+\eta(K, \delta)}(z), \quad (9.4)$$

where

$$\eta(K, \delta) \leq C_{\text{Br}} \left(K^{-2/3} + C_K \sqrt{\delta} \right)^{\mu} + C_K \sqrt{\delta}. \quad (9.5)$$

Here $\mu > 0$ is Brandolini's dimensional exponent and C_{Br} depends only on the selected regularity constants from (SR).

Proof. Set $v = u - \hat{t} = u_{E_{\hat{t}}}$. In dimension two, $w = -2v$ satisfies $\Delta w = 2$ in $E_{\hat{t}}$ and $w = 0$ on $\partial E_{\hat{t}}$. By (8.3),

$$\| |\nabla v| - \bar{f} \|_{L^{\infty}(\partial E_{\hat{t}})} \leq CK^{-2/3}. \quad (9.6)$$

The flux identity gives

$$\bar{f} = \frac{|E_{\hat{t}}|}{P(E_{\hat{t}})}. \quad (9.7)$$

Since the good level has $|E_{\hat{t}}| = \pi + O_K(\sqrt{\delta})$ and $P(E_{\hat{t}}) = 2\pi + O_K(\sqrt{\delta})$, we have

$$\left| \bar{f} - \frac{1}{2} \right| \leq C_K \sqrt{\delta}. \quad (9.8)$$

Since $|Dw| = 2|\nabla v|$, (9.6) and (9.8) give

$$\| |Dw| - 1 \|_{L^{\infty}(\partial E_{\hat{t}})} \leq C \left(K^{-2/3} + C_K \sqrt{\delta} \right). \quad (9.9)$$

Lemma 9.1 supplies connectedness, diameter control, and the uniform $C^{2,\alpha}$ atlas constants required in Brandolini's Theorem 2. Therefore Brandolini gives concentric inner and outer balls for $E_{\hat{t}}$ with gap bounded by the first term in (9.5). The final area normalization changes all radii by $1 + O_K(\sqrt{\delta})$, which adds the last term in (9.5). $\square$

Thus the annular core needed by the Plan 3 radial-hull pocket lemma is indeed available from the selection regularity route: first choose K so large that the $K^{-2/3}$ contribution is below the Brandolini/graph threshold, and then choose δ so small that the $C_K \sqrt{\delta}$ errors are smaller than the same threshold.

Remark 9.3. *This check does not use BDV's final stability theorem as a black box. It uses only the selected-minimizer regularity package and Brandolini's Serrin stability theorem. If the available selection output were only Lipschitz boundary, density, and bounded $|\nabla u|$, then the argument would fail exactly at Lemma 9.1: Brandolini's Theorem 2 has a constant depending on the $C^{2,\alpha}$ atlas of the domain.*

10 Verdict

The user's split patches the scale objection. The estimate

$$\| |\nabla u| - \bar{f} \|_{L^{\infty}(\partial E_{\hat{t}})} \leq C \left(\frac{\delta}{\lambda^2} \right)^{1/3}$$

forces the $\sqrt{\delta}$ -inset scale, but this is compatible with the proof by contradiction because the only nontrivial case has $\alpha(U) \gg \sqrt{\delta}$.

Thus the remaining issue is not a scale theorem. If the selected regularity input (SR) is available, Brandolini supplies the same-center annular core for the stopped level. Combined with the Plan 3 radial-hull pocket lemma, this gives the required radialization at $O(\delta)$ fill cost and closes the planar hybrid route.

If (SR) is weakened to only Lipschitz boundary, density, and bounded $|\nabla u|$, then the proof stops exactly at Brandolini's regularity constant: one has no uniform $C^{2,\alpha}$ atlas for the stopped level.